

Weighted Support Criteria for Reciprocal Mersenne Subseries

The series $\sum_{a \in A} (b^a - 1)^{-1}$ and Erdős Problem #257

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ABSTRACT

For a finite nonempty prime set P , put $h(a) = \prod_{p \in P} p^{v_p(a)}$. We prove that every infinite A with

$$\sum_{a \in A} \frac{h(a)}{a(2^{h(a)} - 1)} < \infty$$

has irrational $\sum_{a \in A} (b^a - 1)^{-1}$ at every integer base $b \geq 2$. The condition is hereditary and strictly weaker than reciprocal summability: an explicit host has divergent $\sum_{a \in A} 1/a$ while satisfying the weighted condition. Finite gcd-orbit averages and a second average over dyadic observation lengths produce arbitrarily small positive displacements, which rationality would confine to a fixed lattice. Erdős already stated the weaker coprimality-free reciprocal-summable theorem, including its all-base quantifier; a complete proof is printed here as well. A uniform finite averaging estimate also combines the weighted criterion with positive divisor covers: their union has the same hereditary irrationality property. These results have ordinary proofs; their analytic steps are not end-to-end Lean theorems. Universal irrationality remains unresolved.

1 Introduction and main results

Write $X_A(b) = \sum_{a \in A} (b^a - 1)^{-1}$ for $A \subseteq \mathbb{N}_{>0}$ and an integer $b \geq 2$. We begin with the simpler reciprocal-summable case; Section 3 proves the stronger finite-prime weighted criterion and its combination with positive divisor covers.

Theorem 1.1 (reciprocal-summable supports). *Let $A \subseteq \mathbb{N}_{>0}$ be infinite. If*

$$\sum_{a \in A} \frac{1}{a} < \infty,$$

then $X_A(b)$ is irrational for every integer $b \geq 2$.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

The statement is ordinary mathematics in this note. This checkout has no module `AllBaseReciprocalSupportIrrationality.lean` and no Lean declaration `irrational_erdosSupportSeries_of_summable_reciprocal`. The nearby [pairwise-coprime support theorem](#) keeps an extra pairwise-coprimality hypothesis together with summable reciprocal mass, and is therefore a structured-support theorem, not Theorem 1.1. Full support at every integer base is the separate [full-support irrationality theorem](#). Arbitrary infinite support remains open.

For infinite A and $N > 0$, division of N by each $a \in A$ gives

$$0 < \Delta_{b,A}(N) := \sum_{a \in A} \frac{b^{N \bmod a} - 1}{b^a - 1} = (b^N - 1)X_A(b) - J_{b,A}(N), \quad J_{b,A}(N) \in \mathbb{Z}. \quad (\text{D})$$

Only $a \leq N$ contribute to $J_{b,A}(N)$. Every summand in the displacement is nonnegative, and an exponent $a > N$ gives a positive summand; such an exponent exists because A is infinite. Rationality $X_A(b) = p/q$ would force every positive displacement to be at least $1/q$. The proof makes these positive displacements arbitrarily small by averaging along multiples of a growing divisibility modulus.

The conclusion is hereditary: every infinite subset of such an A satisfies the theorem at every base. For example, every powerful integer can be written as u^2v^3 , so

$$\sum_{\substack{a \geq 1 \\ a \text{ powerful}}} \frac{1}{a} \leq \zeta(2)\zeta(3) < \infty.$$

Thus arbitrary infinite thinnings of the powerful integers are included.

Problem 1.2 (Erdős #257). Is $X_A(2)$ irrational for every infinite $A \subseteq \mathbb{N}_{>0}$?

Theorem 1.1 excludes the entire reciprocal-summable region. Theorem 3.1 reaches a strictly larger class and an explicit reciprocal-divergent host. Theorem 3.3 combines weighted supports and positive covers by producing a common small displacement. Later sections keep the quantitative, prime-power and stronger host constructions separate where their arguments are not independently assessable from this checkout. The final section retains the actual arithmetic constraint for a proposed rational value.

Erdős proved the pairwise-coprime case at every integer base and stated that the coprimality condition could be removed, leaving the details unprinted [5, p. 222]. Theorem 1.1 gives a complete proof of that stated extension. The all-base quantifier is already in Erdős's statement.

The coefficient sequence throughout is the divisor transform of one Boolean selector:

$$c_A(n) = \#\{a \in A : a \mid n\}, \quad X_A(b) = \sum_{n \geq 1} \frac{c_A(n)}{b^n}. \quad (1)$$

Nonnegative interchange proves this identity. The coefficients may exceed one, and their common origin in a single support will matter below.

2 Reciprocal-summable supports at every integer base

The proof freezes a finite portion of the support and averages the rest. Making the shift divisible by the frozen exponents makes their displacement exactly zero. The remaining exponents need not have a common period of manageable size: reciprocal summability bounds their total average directly.

Fix the integer base $b \geq 2$ and put

$$w_{b,d}(N) = \frac{b^{N \bmod d}}{b^d - 1}, \quad T_N^{(b)} = \sum_{d \in A} w_{b,d}(N).$$

Then $T_N^{(b)} - T_0^{(b)} = \Delta_{b,A}(N)$. For a fixed positive integer Q , the orbit of Q modulo d consists of d/g residues, where $g = (Q, d)$. Summing the geometric progression gives

$$M_{Q,b}(d) := \lim_{X \rightarrow \infty} \frac{1}{X} \sum_{m=1}^X w_{b,d}(Qm) = \frac{g}{d(b^g - 1)} \leq \frac{1}{d}. \quad (2)$$

For example, at $b = 2$, $Q = 4$ and $d = 6$, the orbit is $4, 2, 0$ modulo 6. Its mean atom is $(16 + 4 + 1)/(3 \cdot 63) = 1/9$, whereas its zero-shift atom is $1/63$. If $6 \mid Q$, the orbit instead stays at zero and the mean is exactly $1/63$. This is the effect of freezing an exponent; the infinite tail still requires the bound that follows. The passage from individual atoms to their infinite sum needs a uniform bound. Counting positive multiples of d gives

$$\begin{aligned} \frac{1}{X} \sum_{m=1}^X w_{b,d}(Qm) &= \sum_{r \geq 1} b^{-r} \frac{\#\{1 \leq m \leq X : d \mid Qm + r\}}{X} \\ &\leq \sum_{r \geq 1} b^{-r} \frac{Q + r}{d} = \frac{Q/(b-1) + b/(b-1)^2}{d} \leq \frac{Q+2}{d}. \end{aligned}$$

Indeed, the counted multiples are distinct and at most $QX + r$. Reciprocal summability now permits dominated convergence, so the Cesaro mean of $T_{Qm}^{(b)}$ tends to $\sum_{d \in A} M_{Q,b}(d)$.

Next let $Q_t = \text{lcm}(1, \dots, t)$. For each fixed d , eventually $d \mid Q_t$ and $M_{Q_t,b}(d) = (b^d - 1)^{-1}$. A second dominated-convergence argument, using $M_{Q_t,b}(d) \leq 1/d$, yields

$$\sum_{d \in A} M_{Q_t,b}(d) \rightarrow X_A(b). \quad (3)$$

Thus the limiting averages of the nonnegative displacements $\Delta_{b,A}(Q_t m)$ tend to zero. Choose t , then a sufficiently long finite average, and finally a term no larger than that average. This gives arbitrarily small positive displacements, contradicting the rational lattice in (D). This proves Theorem 1.1 directly at every integer base.

The order of limits is essential: the observation length tends to infinity with the modulus fixed, and only then does the modulus increase. Reciprocal summability controls both interchanges.

Section 2 contains the complete reciprocal-summable support argument. The next section proves a strictly weaker finite-prime weighted hypothesis. The later quantitative, prime-power and cost-separation claims remain separately unassessed where their proofs are not printed here.

3 Extensions beyond reciprocal summability

The finite-prime weighted theorem below has a complete ordinary proof in this manuscript. The finite estimate (S) and strengthened cover theorem later in this section retain their separate proofs.

3.1 Finite prime-part weighted mass

For a finite nonempty prime set P , put $h(a) = \prod_{p \in P} p^{v_p(a)}$.

Theorem 3.1 (finite prime-part weighted mass). *Let $b \geq 2$ be an integer and let $A \subseteq \mathbb{N}_{>0}$ be infinite. If*

$$W_{b,P}(A) := \sum_{a \in A} \frac{h(a)}{a(b^{h(a)} - 1)} < \infty, \quad (4)$$

then $X_A(b)$ is irrational. In particular, the base-two condition

$$\sum_{a \in A} \frac{h(a)}{a(2^{h(a)} - 1)} < \infty \quad (W)$$

implies that $X_A(b)$ is irrational for every integer $b \geq 2$. Both conclusions are hereditary under passage to infinite subsets.

Proof. The rational-lattice obstruction is (D): if $X_A(b) = p/q$, every positive $\Delta_{b,A}(m)$ is at least $1/q$. For $d_a(m) = (b^{m \bmod a} - 1)/(b^a - 1)$ and $g = (Q, a)$, one complete orbit of $Qm \bmod a$ gives, for $Q, a, T \geq 1$,

$$\frac{1}{T} \sum_{t=1}^T d_a(tQ) \leq \frac{g}{a(b^g - 1)} + \frac{1}{T(b^g - 1)}. \quad (5)$$

Indeed, the orbit has length a/g and $\sum_{t=1}^{a/g} b^{tQ \bmod a} / (b^a - 1) = 1/(b^g - 1)$; an incomplete orbit costs at most one more complete orbit. Also, for $Y = QT$,

$$\frac{1}{T} \sum_{t=1}^T \sum_{\substack{a \in A \\ a > Y}} d_a(tQ) \leq \frac{4}{T}, \quad (6)$$

because $d_a(tQ) \leq 22^{tQ-a}$ when $a > QT$ and $\sum_{t=1}^T 2^{tQ-QT} \leq 2$.

The incomplete-orbit errors are controlled by a second finite average. For $\alpha_a \geq 0$ with $\sum_a \alpha_a/a < \infty$,

$$\sum_{j=M}^{2M-1} \frac{1}{2^j} \sum_{a \leq Q2^j} \alpha_a \leq 2Q \sum_{a \geq 1} \frac{\alpha_a}{a}; \quad (7)$$

for each fixed a , the admissible geometric tail is at most $2Q/a$.

Fix $\varepsilon > 0$. Choose finite nonempty $F \subseteq A$ so that the (4) mass outside F is below ε , and let L be any fixed positive common multiple of F . With $p_* = \max P$, $r = |P|$, and large $H \geq 2p_*$, set

$$Q = L \prod_{p \in P} p^{\lfloor \log_p H \rfloor}, \quad G = \left\lfloor \frac{H}{p_*} \right\rfloor.$$

The set F is frozen. If $h(a) \leq H$, then $h(a) \mid Q$, so (5) is paid by the weighted tail plus its incomplete-orbit term. If $h(a) > H$, then $(Q, a) \geq G$, and monotonicity of $n/(b^n - 1)$ bounds that part by

$$\frac{G(1 + \log(QT)) + Q}{b^G - 1}.$$

For the gcd claim, either every P -prime-power component of $h(a)$ is at most H , in which case $h(a) \mid Q$, or some $p^{v_p(a)} > H$ contributes $p^{\lfloor \log_p H \rfloor} > H/p \geq H/p_* \geq G$ to the gcd. Combining this with (6), taking $M = \lfloor b^{G/2} \rfloor$, averaging over $T = 2^j$ for $M \leq j < 2M$, and using (7) with $\alpha_a = \mathbf{1}_A(a)/(b^{h(a)} - 1)$ yields

$$\frac{1}{M} \sum_{j=M}^{2M-1} \frac{1}{2^j} \sum_{t=1}^{2^j} \Delta_{b,A}(tQ) \leq \varepsilon + \frac{2QW_{b,P}(A)}{M} + \frac{G(1 + \log Q + 2M \log 2) + Q}{b^G - 1} + 4 \cdot 2^{-M}. \quad (8)$$

Here $Q \leq LH^r$, $G = H/p_* + O(1)$ and $M \asymp b^{G/2}$, so every term after ε tends to zero. The left side is a finite average of positive displacements, so one is below 2ε for large H ; rerunning with $\varepsilon/2$ gives the required bound. This contradicts the rational lattice. Finally $W_{b,P}(A) \leq W_{2,P}(A)$ for $b \geq 2$, and weighted mass decreases on taking subsets. $\square$

The hypothesis is genuinely weaker than reciprocal summability. Let

$$A_* = \{2^k m : k \geq 1, m \text{ odd}, m \leq 2^{2^k}\}.$$

If $N_k = 2^{2^k}$, the reciprocal mass in its k th layer is

$$R_k = 2^{-k} \sum_{\substack{m \leq N_k \\ m \text{ odd}}} \frac{1}{m} = 2^{-k} (H_{N_k} - \frac{1}{2} H_{N_k/2}) \rightarrow \frac{\log 2}{2},$$

so $\sum_{a \in A_*} 1/a$ diverges. For $P = \{2\}$ the weighted mass of that layer is

$$\frac{2^k}{2^{2^k} - 1} R_k,$$

and these terms are summable. Thus every infinite subset of A_* has irrational $X_A(b)$ at every integer base. This example establishes strictness only; the stronger previously proposed host with $O(\log \log a)$ gaps is not asserted here.

The theorem and this example are complete ordinary proofs. No single Lean declaration is cited as their proof, and no priority claim is made. The arbitrary-chain, quantitative and prime-power extensions remain separate from Theorem 3.1.

3.2 Claimed quantitative criterion (unassessed here)

The absent analytic supplement claims a second criterion admitting reciprocal-divergent supports. Put $H_A(x) = \sum_{a \in A, a \leq x} 1/a$, $T_0 = 2$, $T_{j+1} = 2^{T_j}$, and $\ell(x) = \min\{j : x \leq T_j\}$. The condition $H_A(x) = o(\ell(x))$ is claimed to imply all-base irrationality, with no positive density assumption on A . Neither a proof of this implication nor the claimed sharper rational-phase lower bound is present in this checkout.

3.3 A finite estimate on common observation scales

For $1 < B \leq 2$, positive integers L, d, M , and an integer $R \geq 0$, put

$$w_{B,d}(n) = \frac{B^{n \bmod d}}{B^d - 1}, \quad \mathcal{D}_{L;R,M} F = \frac{1}{M} \sum_{j=R}^{R+M-1} \frac{1}{2^j} \sum_{m=1}^{2^j} F(Lm).$$

The finite estimate

$$\mathcal{D}_{L;R,M} w_{B,d} \leq \frac{1 + 4L/M}{d(B-1)} \quad (\text{S})$$

is uniform in d, R , and B as B decreases to one. The ratio L/M measures the cost of incomplete modular periods.

To prove it, fix $T = 2^j$ and average over $1 \leq m \leq T$. When $d \leq LT$, the complete orbit has length d/g , with $g = \langle L, d \rangle$, and total weight $1/(B^g - 1)$. Complete cycles and one remaining piece give

$$\frac{1}{T} \sum_{m=1}^T w_{B,d}(Lm) \leq \frac{g}{d(B^g - 1)} + \frac{1}{T(B^g - 1)} \leq \frac{1}{d(B-1)} + \frac{1}{T(B-1)}.$$

Across dyadic lengths satisfying $d \leq L2^j$, the reciprocal-length errors sum to at most $2L/[d(B-1)]$. When $d > 2LT$, there is no wrap and $2Lm \leq d-1$; hence $\sum_{i=0}^{d-1} B^i \geq dB^{(d-1)/2} \geq dB^{Lm}$. Each atom is then at most $1/[d(B-1)]$. Finally, at most one dyadic length satisfies $LT < d \leq 2LT$. For that length the geometric sum and convexity give

$$\frac{1}{T} \sum_{m=1}^T w_{B,d}(Lm) = \frac{B^L}{T(B^L - 1)} \frac{B^{LT} - 1}{B^d - 1} \leq \frac{LB^L}{d(B^L - 1)} \leq \frac{2L}{d(B-1)}.$$

Summing the main terms and these two error budgets proves (S). The complete-cycle mass $1/(B-1)$, the bound $g/(B^g - 1) \leq 1/(B-1)$ used for complete cycles, and the no-wrap inequality $\sum_{i < d} B^i \geq dB^{(d-1)/2}$ are ordinary in this note; PaperCompleter7/CoverKernel.lean is not in this checkout. Nonnegative interchange permits summation against any coefficients c_d with $\sum_d c_d/d < \infty$.

3.4 Positive fractional divisor covers

For a finite set F , put $f_F(n) = \#\{a \in F : a \mid n\}$. Let F_j be finite sets of positive integers, let $0 < \alpha_j \leq 1$, and suppose $c_{j,d} \geq 0$ satisfy

$$f_{F_j}(n)^{\alpha_j} \leq \sum_{d \mid n} c_{j,d}, \quad C_j = \sum_{d \geq 1} \frac{c_{j,d}}{d}.$$

Theorem 3.2 (strengthened positive covers). *If*

$$\sum_{j \geq 1} \frac{C_j 2^{j\alpha_j}}{2^{\alpha_j} - 1} < \infty, \quad (\text{V})$$

then $X_A(b)$ is irrational for every infinite $A \subseteq \bigcup_j F_j$ and every integer $b \geq 2$.

Proof. Write $B_j = 2^{\alpha_j}$ and

$$U_j(N) = \sum_{r \geq 1} 2^{-r} f_{F_j}(N+r), \quad V_j(N) = \sum_{d \geq 1} c_{j,d} w_{B_j,d}(N).$$

Subadditivity of the fractional powers and the positive majorants give $U_j(N)^{\alpha_j} \leq V_j(N)$. The geometric-series identity also gives

$$\Delta_{2,F_j}(N) = U_j(N) - X_{F_j}(2) \leq U_j(N).$$

Consequently, once the first J frames have zero displacement, nonnegativity bounds the displacement of their union by $\sum_{j>J} U_j(N)$.

Fix $\varepsilon > 0$, set $t_j = \varepsilon 2^{-j}$, and choose J with

$$K_J := \sum_{j>J} \frac{C_j t_j^{-\alpha_j}}{B_j - 1} < \frac{1}{4}.$$

This is possible since $\varepsilon^{-\alpha_j} \leq \max(1, \varepsilon^{-1})$. Choose L divisible by every member of the first J frames. Equation (S) bounds the finite mean of $S_J(N) := \sum_{j>J} t_j^{-\alpha_j} V_j(N)$ by $(1 + 4L/M)K_J < 1/2$ whenever $M \geq 4L$. One sample therefore has $S_J(N) < 1$, forcing $U_j(N) < t_j$ for every $j > J$. The frozen frames have zero displacement, so

$$0 < \Delta_{2,A}(N) \leq \sum_{j>J} U_j(N) \leq \varepsilon.$$

The fixed rational lattice excludes rationality at base two. For every $b \geq 2$, the elementary atom inequality $(b^r - 1)/(b^d - 1) \leq 2(2^r - 1)/(2^d - 1)$ for $0 \leq r < d$ gives the same conclusion at base b ; that inequality is ordinary in this note (CoverKernel.lean is absent). Any prescribed positive divisor can be included in L , so the witnesses can also be required to be arbitrarily large. $\square$

The same proof permits any positive weights η_j with $\sum_j \eta_j = 1$: replace $2^{j\alpha_j}$ in (V) by $\eta_j^{-\alpha_j}$ and take $t_j = \varepsilon \eta_j$. The condition strengthens the earlier two-inverse-power cost $\sum_j C_j 2^{j\alpha_j} 2^{\alpha_j} / (2^{\alpha_j} - 1)^2 < \infty$. The unavailable *Strengthened Variable-Exponent Cover* supplement claims a squarefree support A^* satisfying (V) for which every cover with that older cost, and every finite-prime weighted criterion, fails. Its construction is not in this checkout and is not a result of the present release.

3.5 What every positive cover must pay

The strengthening has a cover-independent boundary. For finite F , write $\mathbb{E}_F$ for the uniform mean modulo $\text{lcm}(F)$. More generally allow weights $\eta_j > 0$ with $\sum_j \eta_j = 1$, and set

$$K = \sum_j \frac{C_j \eta_j^{-\alpha_j}}{2^{\alpha_j} - 1}, \quad \Psi(0) = 0, \quad \Psi(t) = \inf_{0 < \alpha \leq 1} \frac{t^\alpha}{2^\alpha - 1} \quad (t \geq 1).$$

For every finite F covered by the frames,

$$K \geq \mathbb{E}_F \Psi(f_F) \geq e \mathbb{E}_F \log^+ f_F. \quad (9)$$

Indeed, at a point where $f_F(n) = t > 0$, some frame has $f_{F_j}(n) \geq \eta_j t$. Otherwise summing contradicts coverage. The corresponding weighted majorant is at least $\Psi(t)$. Averaging and using $\lfloor X/d \rfloor / X \leq 1/d$ proves the first inequality. Convexity gives $2^\alpha - 1 \leq \alpha$, and $e^u / u \geq e$ proves the second. Those two scalar steps, and the bound $t^\alpha / (2^\alpha - 1) \geq e \log t$ at every admissible exponent, are ordinary in this note (CoverKernel.lean is absent).

This bound survives optimisation over all covers. For $F(q, P) = \{qd : d \mid \prod_{p \in P} p\}$, where $q \geq 2$ and no $p \in P$ divides q , put $S = \sum_{p \in P} 1/p$. If $S \geq 1$, the infimum K_* over all covers satisfies

$$\frac{e(S-1)}{q} \leq K_*(F(q, P)) \leq \frac{eS}{q}. \quad (10)$$

For the lower bound, condition on $q \mid n$ and write $f_F(n) = 2^Z$. The Chinese remainder theorem gives $\mathbb{E}Z = S$, and direct minimisation gives $\Psi(2^z) \geq e(z-1)$. For the upper bound, use the single frame, $z = 1/S$, and $\alpha = \log_2(1+z)$; its exact positive expansion has cost

$$\frac{1}{qz} \prod_{p \in P} (1 + z/p) \leq \frac{eS}{q}.$$

Thus the optimised cost is asymptotically eS/q .

An unavailable supplement claims that the weighted class and the strengthened-cover class are incomparable. Its proposed support A^* lies only in the latter. In the reverse direction, it proposes disjoint prime blocks with reciprocal mass of order 2^k , placed in frames $\{2^k d : d \mid M_k\}$, to give a support A_W satisfying (W) whose first logarithmic moments diverge; then (9) would exclude every strengthened cover. The exact constructions are referred to *First Logarithmic Moment of Positive Covers*, which is not in this checkout, so this incomparability is not a result of the present release.

3.6 Combining the two support criteria

Separate small-displacement witnesses need not occur at the same index. For example, a sequence small only at even indices and one small only at odd indices need never have a small sum. The useful feature of (S) is its uniformity in the moving modulus: the positive-cover argument can use the exact observation window selected by the weighted proof.

Theorem 3.3 (mixed weighted and cover supports). *Suppose E has finite weighted mass (W) for a finite nonempty prime set P , and V admits a strengthened positive cover (V), or its positive-weight variant above. Then $X_A(b)$ is irrational for every infinite $A \subseteq E \cup V$ and every integer $b \geq 2$.*

Proof. Fix $\varepsilon > 0$ and an integer $N_0 \geq 1$, and put $\rho = \varepsilon/3$. Use the cover notation B_j, U_j, V_j above, with $\eta_j = 2^{-j}$ in the case (V). Set $t_j = \rho \eta_j$ and choose J so that

$$K_J = \sum_{j > J} \frac{C_j t_j^{-\alpha_j}}{B_j - 1} < \frac{1}{16}.$$

Choose a finite $F \subseteq E$ whose complementary weighted mass is $\kappa < \rho/16$. Let L be a positive common multiple of N_0 , all members of F , and all members of the first J cover

frames. For large H , take exactly the binary weighted schedule

$$Q = L \prod_{p \in P} p^{\lfloor \log_p H \rfloor}, \quad G = \lfloor H / \max P \rfloor, \quad M = \lfloor 2^{G/2} \rfloor.$$

The finite estimate (8), with tail budget κ , remains valid for this L : its proof requires only that L freeze F . It also applies to finite or empty E , since positivity was used only after the averaging estimate. Thus

$$\mathcal{D}_{Q;M,M}(\Delta_{2,E}/\rho) < \frac{1}{8}$$

for sufficiently large H . For the same finite distribution, (S) and nonnegative interchange give

$$\mathcal{D}_{Q;M,M} S_J \leq (1 + 4Q/M) K_J < \frac{1}{8}, \quad S_J = \sum_{j>J} t_j^{-\alpha_j} V_j,$$

because $Q/M \rightarrow 0$. Hence some sample $N = Qm$ satisfies $\Delta_{2,E}(N)/\rho + S_J(N) < 1$. At this one index the weighted displacement is below ρ and every unfrozen cover frame has $U_j(N) < t_j$. Therefore

$$\Delta_{2,A}(N) \leq \Delta_{2,E}(N) + \Delta_{2,V}(N) < 2\rho < \varepsilon, \quad N \geq Q \geq L \geq N_0.$$

Overlaps between the supports only improve the inequality. Infinitude of A makes its displacement positive. The atom comparison used in the cover proof transfers arbitrarily small displacements to every integer base; the fixed rational lattice then excludes rationality. $\square$

This is a complete ordinary proof. It does not establish the unavailable constructions A_W, A^* or strict enlargement over both individual classes. Those separation claims retain the boundary stated above.

With arbitrary positive cover weights, the class of subsets of such mixed hosts is closed under finite unions and finite changes. For weighted supports use the union of the two finite prime sets: the prime part grows and $h/(2^h - 1)$ decreases. For two covers, interleave their frames with weights $\eta_j/2$ and $\theta_j/2$; the total cost grows by at most two, since $2^{\alpha_j} \leq 2$. Subsets inherit the same hosts, and finite sets have finite weighted mass. This argument uses the positive-weight variant; it does not silently reindex the dyadic weights in (V). Countable unions require a tail budget. Every prime singleton is admitted, but for the full prime support $\mathcal{D}$ one has $\Delta_{2,\mathcal{D}}(N) > 1/3$ for every $N \geq 1$: indeed $\sum_{r \geq 1} 2^{-r} \omega(N+r) \geq 1$, whereas $X_{\mathcal{D}}(2) \leq \sum_{a \geq 2} (2^a - 1)^{-1} < 2/3$. Here $\omega(n)$ is the number of distinct prime divisors of n . The reciprocal-summable class is contained in the weighted class, since $h/(2^h - 1) \leq 1$. Theorem 1.1 supplies the direct proof of that baseline case.

The prime-power regime. A further claimed extension would give irrationality for every infinite subset of the prime powers, at every integer base, including fixed dilations and finite modifications. Its intended analytic input is Tao–Teräväinen [2, Theorem 3.1].

The argument would use the reciprocal mass of the selected primes as its scale, so arbitrarily slow divergence is retained; higher prime powers produce errors only on prime-square events. The equidistribution, small-prime, progression and exceptional-set hypotheses are referred to a supplement that is not in this checkout. The full-prime theorem and the authors' stated full-prime-power extension are theirs. No arbitrary thinning argument is promoted from this release.

The actual greedy membership problem is treated in Section 8; its repair inequality has a different logical role from the sufficient irrationality criteria above.

4 Finite-support denominator periods

For a finite nonempty $F \subseteq \mathbb{N}_{>0}$, let D_F be the positive reduced denominator of $x_F(b) = \sum_{n \in F} (b^n - 1)^{-1}$. We use $\text{ord}_1(b) = 1$.

Theorem 4.1 (finite-period noncollapse). *Let $F \subseteq \mathbb{N}_{>0}$ be finite and nonempty, let $b \geq 2$ be an integer, and let $D_F > 0$ be the denominator of $x_F(b)$ in lowest terms. Then D_F is coprime to b , and*

$$\text{ord}_{D_F}(b) = \text{lcm}\{n : n \in F\}.$$

If moreover $\text{lcm}(F) \geq 2$, then $\text{lcm}(F) < D_F$.

Coprimality is [base coprimality of the reduced denominator](#), the order statement is [the exact multiplicative order](#), and the size bound is [the strict denominator inequality](#). The three clauses are the CertificateKernel declarations just cited; PaperCompleteR7/Assemblies.lean is not in this checkout.

Put $L = \text{lcm}(F)$. Clearing denominators gives $D_F \mid b^L - 1$, so the upper divisibility for the order is immediate. For the reverse, choose $n \geq 2$ maximal under divisibility in F and a prime $\ell \mid \Phi_n(b)$. If $e = v_\ell(b^n - 1)$, the full prime power ℓ^e has $\text{ord}_{\ell^e}(b) = n$. Every other selected exponent m has $n \nmid m$, hence $v_\ell(b^m - 1) < e$. The n th summand has uniquely smallest ℓ -adic valuation and cannot cancel. Thus $\ell^e \mid D_F$ and $n \mid \text{ord}_{D_F}(b)$. Taking all maximal selected exponents proves the order statement; the size bound follows from $\text{ord}_{D_F}(b) \mid \varphi(D_F) < D_F$ when $L \geq 2$. The case $F = \{1\}$ has order one directly.

The same unique-valuation argument permits any signs ± 1 on the finite summands; that signed extension is ordinary in this note (SignedFinitePeriodNoncollapse.lean is not in this checkout). Its cyclotomic prime-power lemma, including the 2-adic case, is proved in *Finite Denominator Realisation*. The distinction between primes and prime powers is visible in

$$X_{\{2,3\}}(2) = \frac{10}{21}, \quad X_{\{2,6\}}(2) = \frac{22}{63}.$$

Both denominators have order six. The first combines orders two and three, while the second retains the order-six prime power 9; no prime divisor of 63 itself has order six. The boundary $F = \{1\}$ at base two has $D_F = L = 1$. These lower denominator bounds give no upper height control for infinite partial sums and do not decide an infinite-support value.

5 Rational values and scaled tails

The displacement identity gives a fixed arithmetic lattice under rationality. Its recurrence is

$$z_{N+1} = 2z_N - vc_A(N+1), \quad z_N = v \sum_{r \geq 1} c_A(N+r)2^{-r} \in \mathbb{Z}$$

when $X_A(2) = p/v$. The sequence c_A is determined by one Boolean selector.

Two elementary facts explain why size information alone has little force here. If $a_0 = \min A$, every a_0 consecutive integers contain a multiple of a_0 , so a divisor-incidence zero window has length at most $a_0 - 1$. If A is infinite, choose k exponents and a common multiple L ; then $c_A(L) \geq k$ and $T_{L-1}^{(2)} \geq k/2$. Both facts hold without rationality. The arithmetic information lies in the lattice and in the compatibility of all coefficients with the same selector.

Precisely, Dirichlet convolution gives $\mu * c_A = \mathbf{1}_A$, where μ is the Möbius function. A putative recurrence with integral forcing must therefore satisfy $(\mu * c_A)(n) \in \{0, 1\}$ for all n . The complete Boolean–Möbius scaled-tail correspondence and its finite example are retained in the long record, under *Exact descriptions of a rational-valued support* (Theorem thm:bmc). The correspondence permits finite supports; infinitude remains a separate requirement for a counterexample to Problem 1.2.

6 Where the return mechanism stops

The preceding criteria establish irrationality through small positive returns. This mechanism already fails at the full support:

$$\Delta_{2, \mathbb{N}_{>0}}(N) > (2^N - 1) \sum_{a > N} 2^{-a} = 1 - 2^{-N} \geq \frac{1}{2} \quad (N \geq 1).$$

The full-support value is nevertheless [irrational](#) [5]. A universal proof must therefore control arithmetic behaviour beyond small returns.

The distinction also appears in the squarefree support. Its series is known to be irrational at powers-of-two bases by Duverney–Tachiya [1, Corollary 1.2 and Example 1.1]. The zero-window obstruction for one normalised certificate scheme concerns that scheme’s hypotheses. It gives no obstruction to irrationality of the value. The precise normalisation counterexamples are retained in the long record, together with the complete catalogue of known support families.

A related independence boundary concerns prime-incidence arguments. Writing $f_A(n) = \sum_{a \in A} \mathbf{1}_{a|n}$, the local identity

$$f_A(an) - f_A(n) = 1 - \mathbf{1}_{a|n} \quad (n \geq 1)$$

for a fixed $a \in A$ holds exactly when a is coprime to every other member of A . To see this, every summand $\mathbf{1}_{t|an} - \mathbf{1}_{t|n}$ is nonnegative. The term $t = a$ is the displayed right side; all others vanish if $(a, t) = 1$. Conversely, if $g = (a, t) > 1$ for some $t \neq a$, then $n = t/g$ gives an additional positive term, contradicting the identity. Already for the

dilated prime support $2\mathcal{D}$,

$$\text{Cov}(\mathbf{1}_{2p|n}, \mathbf{1}_{2q|n}) = \frac{1}{4pq} \quad (p \neq q \text{ odd primes}),$$

where covariance is taken over a common arithmetic period: the joint mean is $1/(2pq)$ and the product of the means is $1/(4pq)$. Thus multiplicative size conditions alone do not supply the independence used by the prime argument. These are elementary ordinary calculations.

7 Unique coding and arithmetic membership

Let $w_n = (2^n - 1)^{-1}$ and

$$\mathcal{A} = \left\{ \sum_{n \geq 1} \varepsilon_n w_n : \varepsilon_n \in \{0, 1\} \right\}, \quad R_N = \sum_{n > N} w_n.$$

The estimate

$$2^{-N} < R_N \leq 2^{-N} + \frac{2}{3} 4^{-N} < w_N \quad (N \geq 1)$$

shows that every represented value has a unique selector. The middle inequality follows by writing $w_n = 2^{-n} + 4^{-n}/(1 - 2^{-n})$ and using $n \geq N + 1 \geq 2$. The coding image is a Cantor set, and the disjoint level- N cylinder covers have total length $2^N R_N \rightarrow 1$, so $\lambda(\mathcal{A}) = 1$. Kovač-Tao record this strict-tail inequality and the Cantor conclusion in the fixed-base setting [3, Remark 4.1]. The long record proves the restricted-volume dichotomy and records the formula $\dim_H \mathcal{A}_J = \liminf_N \#(J \cap [1, N])/N$, together with the periodic-stride measure refinements. For a specified rational target, the useful consequence here is uniqueness of its possible selector.

8 The actual greedy return inequality

For $x \geq 0$, the greedy rule starts with $r_0 = x$ and, at rank $n \geq 1$, selects n when $r_{n-1} \geq (2^n - 1)^{-1}$, subtracting that weight if selected. Let A_x be the resulting support and $c_x(n) = \#\{a \in A_x : a \mid n\}$. Set

$$P_0 = 0, \quad P_{N+1} = 2P_N + c_x(N+1), \quad Q_N = \lfloor 2^N x \rfloor - P_N.$$

The nonnegative integer defect obeys

$$Q_{N+1} = 2Q_N + \beta_N - c_x(N+1), \quad \beta_N = \lfloor 2^{N+1} x \rfloor - 2\lfloor 2^N x \rfloor \in \{0, 1\}. \quad (11)$$

Theorem 8.1 (general greedy repair criterion). *For every real $x \geq 0$, the following are equivalent:*

$$\begin{aligned} & x \in \mathcal{A}; \\ & \forall K \geq 0 \exists N \geq K : \quad Q_{N+1} \leq Q_N; \\ & \forall K \geq 0 \exists N \in [K, K + 2\lfloor \sqrt{K} \rfloor + 12) : \quad Q_{N+1} \leq Q_N. \end{aligned}$$

The cofinal and windowed equivalences are ordinary in this note; GreedyRepairCriterion.lean and PaperCompleteR7/Assemblies.lean are not in this checkout.

Proof. Strict domination of each Mersenne weight over its tail implies that a represented target is recovered by the greedy rule. For such a target,

$$0 \leq Q_N \leq \sum_{r \geq 1} c_x(N+r)2^{-r} \leq 2\sqrt{N} + 4,$$

since $c_x(n) \leq \tau(n) \leq 2\sqrt{n}$ and $\sqrt{N+r} \leq \sqrt{N} + \sqrt{r} \leq \sqrt{N} + r$. Put $s = \lfloor \sqrt{K} \rfloor$ and $T = 2s + 12$. Strict increase at all T steps would imply $Q_{K+T} \geq T$. But $K + T < (s + 4)^2$ gives $Q_{K+T} < 2s + 12 = T$. This proves the window condition and hence cofinal nonincreases.

Conversely, if $x \notin A$, then $\delta = x - X_{A_x}(2) > 0$. The Lambert prefix is at most $2^N X_{A_x}(2)$, so $Q_N \geq 2^N \delta - 1$. Eventually this exceeds $c_x(N + 1) \leq N + 1$. Equation (11) then makes Q_N strictly increasing, contradicting cofinal nonincreases. $\square$

The explicit square-root estimate is uniform in the target. The ordinary subpower refinement replaces its window length by $\lceil C_\varepsilon(K + 1)^\varepsilon \rceil$ for every $0 < \varepsilon < 1$, using $\tau(n) = O_\delta(n^\delta)$. This changes the permissible deadline without proving any occurrence for a specified target.

At $x = 1/2$, the digits β_N vanish for $N \geq 1$; at $x = 1/21$ they are six-periodic. Thus the unresolved arithmetic input is

$$\forall K \exists N \geq K : \quad c_x(N + 1) \geq Q_N + \beta_N, \quad x \in \{1/2, 1/21\}. \quad (12)$$

Both Q_N and c_x must arise from the same greedy selector. The long record retains exact counterexamples to fixed-multiplier repair schedules and the finite phase masks used to test them. Neither target is decided here.

9 Further questions

The preceding arguments isolate two different tasks. For new irrational supports, one needs a summable or uniformly averaged control of the shifted divisor atoms, or an arithmetic functional that also detects supports whose positive displacements stay separated from zero. For a proposed rational target, one needs the actual-selector inequality (12); synthetic recurrences satisfying only growth and integrality conditions do not provide it.

Theorem 9.1 (fatal-branch quotient refinement at $1/21$). *The following statements hold.*

1. $1/21 \in A$ if and only if $\mathcal{F}_{21}$ does not hold.
2. If there is an unbounded sequence of ranks R with $s_R \leq 2^R$, then $1/21 \in A$.
3. On $\mathcal{F}_{21}$, eventually $s_R > 2^R$, the boundary rank $R + 1$ belongs to D_{R+1} , and (D_R, s_R) follows one exact affine recurrence.

The equivalence is [one div twenty one mem iff not fatal Aligned Branch](#); the closed-row compactness step is [twenty One Cofinal Even Quotient Greedy Decay of closed](#)

Rows; and the eventual affine regime is [twenty One Fatal Aligned Branch eventually affine supercapacity](#). The denominator-specific separation theorem additionally proves that every closed Boolean quotient row is exactly the canonical quotient-greedy row ([twenty One Closed Row forces quotient Greedy](#)). An aligned crossing from saturation into strict supercapacity forces a missing canonical ancestor and a real skipped exponent ([ancestor hole, scaled skip](#)).

Theorem 9.2 (terminal scaled vanishing implies the half-value). *Let S be a HalfTerminalOnlyScaledVanishingSequence: its finite words exclude ranks 0 and 1, their depths tend to infinity, and the absolute terminal carry divided by 2^M tends to zero along the depths M . Then there is an infinite set $A \subseteq \mathbb{N}$ with*

$$\sum_{a \in A} \frac{1}{2^a - 1} = \frac{1}{2}.$$

Consequently the universal irrationality assertion in Problem 1.2 is false under this hypothesis. This is a conditional implication, not a construction of S and not a solution of Problem #257.

Theorem 9.3 (cofinal cylinders imply an infinite half-support). *Suppose that for every N there are M, K with $\max\{N, 1\} \leq M$ and a nonempty CylinderStage $K \leq M$. Then there is an infinite set $A \subseteq \mathbb{N}$ with $0 \notin A$ and*

$$\sum_{a \in A} \frac{1}{2^a - 1} = \frac{1}{2}.$$

Thus the universal irrationality assertion in Problem #257 is false under this cofinal-stage hypothesis. The formal proof does not construct the cofinal family of full-cylinder stages.

Problem 9.4 (membership of $1/21$ in the Mersenne achievement set). Prove cofinal crossings of the exact moving lower separatrix

$$2 \cdot \text{scaledGreedyRemainder}(1/21, N) < \text{mersenneScale}(N + 1).$$

Equivalently, exclude the explicit fatal/cofinite/aligned branch $\mathcal{F}_{21}$. It is sufficient either to contradict the eventual permanent affine-supercapacity recurrence forced by that branch or to force an unbounded sequence of closed canonical quotient rows; neither sufficient route is claimed to be equivalent by itself.

Problem 9.5 (weakest native recurrence criterion). Does the scaled actual greedy remainder return cofinally to one bounded interval?

$$\exists B < \infty \forall K \exists N \geq K : \quad 2^N r_N \leq B.$$

Problem 9.6 (actual-orbit invariant). Is there a finite-memory, 2-adic or discrepancy invariant, using the correlated divisor pulses of the actual support, that forces a closed return or forbids permanent supercapacity? More precisely, can one use a bounded window of $R \bmod 6$, residues of s_R , endpoint divisor counts and the finite set of eventual skips to force descent or a forbidden state? Alternatively, can one prove that no bounded-memory invariant distinguishes the true orbit from synthetic permanent-supercapacity controls?

Problem 9.7 (final-skip Diophantine exclusion). Let $E = \sum_{n \geq 1} (2^n - 1)^{-1}$. If $1/21 \notin A$, let M be the last skipped exponent, S_M its finite skipped prefix, and

$$a_M = \frac{1}{21} + \sum_{d \in S_M} \frac{1}{2^d - 1}.$$

Can the complete final-skip signatures be used to prove $|E - a_M| \geq \text{gap}_M$, contradicting

$$0 < a_M - E < \text{gap}_M?$$

The logarithmic obstruction (9) is necessary for a finite-cost positive cover. Determining when a converse holds would give an intrinsic description of that sufficient class. The mixed class is a finite-union ideal. Countable gluing needs an explicit tail budget: every prime singleton is admitted, while the full prime support has displacement greater than $1/3$ at every positive shift.

Statements and declarations

Finite-period noncollapse has the counterpart [rational-denominator noncollapse](#). Full-support irrationality is [the full-support theorem](#); pairwise-coprime summable-reciprocal support is [the pairwise-coprime theorem](#). Theorem 1.1 itself is ordinary mathematics in this note and has no Lean counterpart in this checkout: the file-names `AllBaseReciprocalSupportIrrationality.lean`, `SignedFinitePeriodNoncollapse.lean` and `GreedyRepairCriterion.lean` are not present. The source checkpoint is 99f4bf47422a; later counterparts are identified separately rather than attributed to that checkpoint.

The finite estimate (S) and Theorem 3.2 have complete ordinary proofs in this note and are results of this release. The finite-prime weighted criterion, Theorem 3.1, and its simple reciprocal-divergent host, the mixed theorem 3.3, and the prime-incidence calculation also have complete ordinary proofs here. They have no end-to-end Lean counterpart in this checkout. The quantitative criterion, the stronger claimed support A^* separating the two cover costs, and prime-power thinning instead refer to ordinary proofs that are not linked in this public checkout (*Strengthened Variable-Exponent Cover*, *First Logarithmic Moment of Positive Covers*, and the analytic proof supplement). Those latter extensions are therefore not independently assessable here and are not promoted as results of this release. No universal irrationality conclusion is asserted.

A Guide to the formal sources

The source annotations in this note deliberately name two immutable revisions. Links produced by the standard note macros use 99f4bf47422a; across this note and its long reasoning surface, fourteen paper-local coordinates link revision f36a98bf3d3e. These families are audited separately; a historical aggregate reference count is not asserted to be an exact inventory of this PDF.

The following source links are the public snapshot used by this note.

[finite period noncollapse](#), [finite period noncollapse rat den](#), [coprime base den finite Erdos Sum](#), [lcm lt den finite Erdos Sum](#). ([binary Coeff Series rational iff exists tempered](#)

Binary Orbit). exists unbounded shifted odd tail nat state of support fraction. support Coeff Zero Window length le eps logb add. one div odd Order le reciprocal Mass of support fraction. dyadic support fraction reciprocal Mass diverges or gt one. exists normalized support fraction iff exists boolean Mobius Carry. irrational erdos Support Series univ

(irrational erdos Support Series multiples)

irrational rat Weight Series eventually Periodic; above (residue class, odd); factorial support

powers-of-two support

factorial-support instance

powers-of-two-support instance

pairwise-coprime support

full-support irrationality; (lcm gap hypothesis fails full support). (irrational or bpow mul eq int Cast int Weighted Erdos Series periodic). card squarefree Divisors, square-free Incidence eq, odd squarefree Incidence. (squarefree Divisors eq image), the carry-aware no-go the digitwise no-go. $2^{\omega(n)}$ incidence the shift iff. for ω for the shifted incidence. irrational erdos Support Series of tail irrational erdos Support Series tail of irrational. compact, perfect, totally disconnected, nowhere dense, of measure one. (mem mersenne Achievement Set iff greedy survival). (three fourths not mem mersenne Achievement Set). (selected Mersenne Tail, summable selected Mersenne Tail). (selected Mersenne Tail It weight), (supported Mersenne Digit Value injective), digit strings vanishing off J restricted digit map, supported digit set, closed. restricted achievement set; the image theorem. compact closed. support restriction nowhere dense. preperfect; A_J , perfect. controls digit terms, the update formula. exactly. disjoint, doubles the volume. by definition. multiplies the face volume by $2^{|F|}$ gives $2^{-|F|}$. supported Mersenne Achievement Set mono. measure zero. the formal dichotomy. the greedy form, the terminal-bit form, the skipped-rank form, the fatal-gap equivalence, its transfer to non-membership, the finite-support exclusion. the local row constructor. the upper-half Boolean fill. the strict-positivity theorem; the cofinal-skip hypothesis, the row fan-in. the closed-set step. forward endpoint theorem the checked equivalence. achievement-set conclusion, infinite-support lift. rational half counterexample, universal-claim refutation. general band localization, two-thirds band, odd numerator bound. integral safety. forced carry supply irrationality endpoint. uniform tail selector. exact dilation identity. prime-support no-defect, prime specialization. foreign-divisor classification defect bound. (two-six witness). terminal-only bridge, infinite half-support, positive-support lift. (identity, bound). the formal theorem. (exists primitive23 solution of eleven le), (no primitive23 solution ten). (rank eleven, multiples of ten). mersenne Achievement Set eq scaled Greedy Trap, scaled Greedy Remainder tendsto at Top of not mem, mem mersenne Achievement Set iff scaled Remainder cofinally Bounded. rat mem mersenne Achievement Set iff scaled Lower Branch Cofinally, one div twenty One mem iff scaled Lower Branch Cofinally. (twenty One Greedy Defect add mod div eq scaled remainder add tail). (support, remainder). one div twenty one mem iff not fatal Aligned Branch; twenty One Cofinal Even Quotient Greedy Decay of closed Rows; twenty One Fatal Aligned Branch eventually affine supercapacity. (twenty

One Closed Row forces quotient Greedy). (ancestor hole, scaled skip). (one div twenty One mem iff scaled Remainder cofinally Bounded). (sufficient condition) (twenty One Greedy Defect add six). the exact equivalence. an exact one-sided approximation. (order identity, lower bound).

References

- [1] D. Duverney and Y. Tachiya, *Refinement of the Chowla–Erdős method and linear independence of certain Lambert series*, Forum Math. 31 (2019), no. 6, 1557–1566, DOI. Corollary 1.2 gives the general $F_s(E)$ theorem and its monomial images under $|q|^{\text{lcm}(1, \dots, \ell)} \leq s$; Example 1.1 gives the joint squarefree family at all bases 2^j , $j \geq 1$. Both are on p. 4 of the linked author preprint; the proof is on pp. 9–11.
- [2] T. Tao and J. Teräväinen, *Quantitative correlations and some problems on prime factors of consecutive integers*, arXiv:2512.01739 (submitted December 2025, revised April 2026). Theorem 1.3 proves $\sum_{n \geq 1} \omega(n)/2^n = \sum_p (2^p - 1)^{-1}$ irrational, settling the prime-support case of #257 at base 2; the extension to every integer base and the prime-power support are stated as remarks with the modifications left to the reader. The theorem is on p. 4 and its proof is Section 5, pp. 44–56, in arXiv v2.
- [3] V. Kovač and T. Tao, *On several irrationality problems for Ahmes series*, Acta Math. Hungar. 175 (2025), 572–608, DOI. Remark 4.1 (p. 13) records the strict tail inequality and the Cantor structure. Theorem 2.3 (p. 5; proof pp. 13–14) constructs rational merged sums from several bases under its mass hypothesis; it is not a fixed-base counterexample.
- [4] P. Erdős, *On arithmetical properties of Lambert series*, J. Indian Math. Soc. 12 (1948), 63–66.
- [5] P. Erdős, *On the irrationality of certain series*, Math. Student 36 (1968), 222–226 (issued 1969). The theorem on p. 222 treats pairwise-coprime support with convergent reciprocal sum at every integer base $b \geq 2$; the claimed removal of pairwise coprimality is stated without proof.

Companion system context. The [claim and trust boundary](#), [cold-clone route to proof authority](#), and [public contribution protocol](#) are described in sibling papers. Those descriptions do not change the mathematical status of this note.